Lab 4 Report: Heart Disease Classification (Neural Network vs. AdaBoost)

Task 1-2: Loading and understanding the data

The UCI Cleveland dataset has **303 samples, 13 features and 1 target**, and every column is numeric (float64, with target as int64). The target target originally takes values 0-4, where 0 means no disease and 1-4 mean increasing severity. The first rows and describe() confirm the data loaded correctly. For example, age ranges from 29 to 77 and cholesterol has an outlier at 564.

Task 3: Class balance

Class	Count	%
0 (no disease)	164	54.13
1 (disease, after binarizing 1-4)	139	45.87

The raw 5-class target is heavily skewed (class 4 has only 13 samples), so I collapsed it into binary. The binary split is **close to balanced**, so no resampling (SMOTE or class weights) was needed.

Task 4: Data quality

- Missing values: **ca has 4 and thal has 2**, 6 in total. Both were marked with ? in the raw file and converted to NaN on loading.
- Duplicate rows: **0**.
- Both columns are categorical-coded and the missing values are under 2% of the data, so I **imputed rather than dropped rows**.

Task 5: Visualizations

- **Target distribution:** this plot confirms the class counts above.
- **Age histogram:** this is roughly bell-shaped, centred in the mid-50s.
- **Cholesterol by target (boxplot):** the classes overlap heavily and there are a few high outliers, so cholesterol alone separates them poorly.
- **Correlation heatmap:** features such as cp, thalach, oldpeak, ca and thal correlate most strongly with the target. thalach (max heart rate) is negatively correlated with it.

Task 6: Preprocessing

- **Numerical features:** age, trestbps, chol, thalach and oldpeak were imputed with the mean and standardized.
- **Categorical features:** sex, cp, fbs, restecg, exang, slope, ca and thal were imputed with the mode and one-hot encoded.
- **Split:** stratified 80/20 train/test, with 15% of the training portion held out for validation.

- **No leakage:** the preprocessor was **fit only on the training data**, and the validation and test sets were only transformed.
- **Result:** one-hot encoding expanded the 13 raw features to **25 input features**.

Task 7: Network design

Layer	Neurons	Activation
Input	25	-
Hidden 1	32	ReLU
Hidden 2	16	ReLU
Hidden 3	8	ReLU
Output	1	Sigmoid (applied inside the loss)

Task 8: Manual parameter count

Formula: (inputs × outputs) + outputs

Layer	Calculation	Parameters
Hidden 1	(25 × 32) + 32	832
Hidden 2	(32 × 16) + 16	528
Hidden 3	(16 × 8) + 8	136
Output	(8 × 1) + 1	9
Total		**1505**

This matches the PyTorch output (1505). **Fix needed:** the notebook's Cell 10 table still uses the placeholder input of 20 (total 1345). Replace it with the table above before submitting.

Task 9: Loss and optimizer

- **Loss: BCEWithLogitsLoss.** The task is binary classification. This loss combines sigmoid and binary cross-entropy in one numerically stable step.
- **Optimizer: Adam (lr = 0.001).** It adapts the learning rate per parameter, converges quickly on small tabular data and needs little tuning.
- **Training:** 100 epochs, batch size 32, with a validation set tracked each epoch.

Task 10: Training vs. validation loss

The model **overfits after about epoch 30**.

- Training loss falls steadily from 0.58 to 0.12.
- Validation loss reaches its minimum of about 0.26 near epoch 30, then rises to 0.55 by epoch 100.

The widening gap means the network is memorizing the training data instead of generalizing. Early stopping around epoch 30 would have avoided this.

Task 11-12: Test results

Model	Accuracy	Precision	Recall	F1
Neural Network	0.8689	0.8333	0.8929	0.8621
AdaBoost (100 estimators)	0.8689	0.8333	0.8929	0.8621

Task 13: Comparison

- **The two models scored identically.** With 61 test patients, these numbers correspond to 25 true positives, 28 true negatives, 5 false positives and 3 false negatives. Identical scores are plausible on a test set this small, where a couple of flipped predictions change the metrics noticeably. The models may still make different individual errors, so it is worth printing both confusion matrices to check.
- **Recall (0.89) is higher than precision (0.83).** The models catch most sick patients at the cost of some false alarms. That trade-off is preferable in a medical setting, because missing a sick patient is worse than a false alarm.
- **Overfitting:** the neural network overfits clearly (see Task 10). It was evaluated after the full 100 epochs, so early stopping could improve its test scores. AdaBoost is simpler and more stable on small data, so it matched the network without any special handling.
- **Preprocessing:** scaling helped the neural network, which is sensitive to feature magnitude. One-hot encoding avoided implying a false order in categorical codes. Fitting the preprocessor on training data only kept the test results honest. AdaBoost is tree-based and does not need scaling.

Task 14: Conclusion

The data is nearly balanced and has very few missing values, so simple imputation was enough. A small 25-32-16-8-1 network and AdaBoost both reach about 87% accuracy and 0.86 F1. The network overfits because the dataset is small. Possible improvements include early stopping, dropout or L2 regularization, k-fold cross-validation for a more reliable estimate, hyperparameter tuning, and a stronger baseline such as XGBoost.

Load data

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	2
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0

```
Shape (rows, columns): (303, 14)

Column names and dtypes:
age         float64
sex         float64
cp          float64
trestbps    float64
chol        float64
fbs         float64
restecg     float64
thalach     float64
exang       float64
oldpeak     float64
slope       float64
ca          float64
thal        float64
target        int64
dtype: object

First 5 rows:
```

First 5 rows:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	2
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0

Descriptive statistics:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	tar
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	299.000000	301.000000	303.000
mean	54.438944	0.679868	3.158416	131.689769	246.693069	0.148515	0.990099	149.607261	0.326733	1.039604	1.600660	0.672241	4.734219	0.937
std	9.038662	0.467299	0.960126	17.599748	51.776918	0.356198	0.994971	22.875003	0.469794	1.161075	0.616226	0.937438	1.939706	1.228
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000	1.000000	0.000000	3.000000	0.000
25%	48.000000	0.000000	3.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	0.000000	1.000000	0.000000	3.000000	0.000
50%	56.000000	1.000000	3.000000	130.000000	241.000000	0.000000	1.000000	153.000000	0.000000	0.800000	2.000000	0.000000	3.000000	0.000
75%	61.000000	1.000000	4.000000	140.000000	275.000000	0.000000	2.000000	166.000000	1.000000	1.600000	2.000000	1.000000	7.000000	2.000
max	77.000000	1.000000	4.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000	3.000000	3.000000	7.000000	4.000

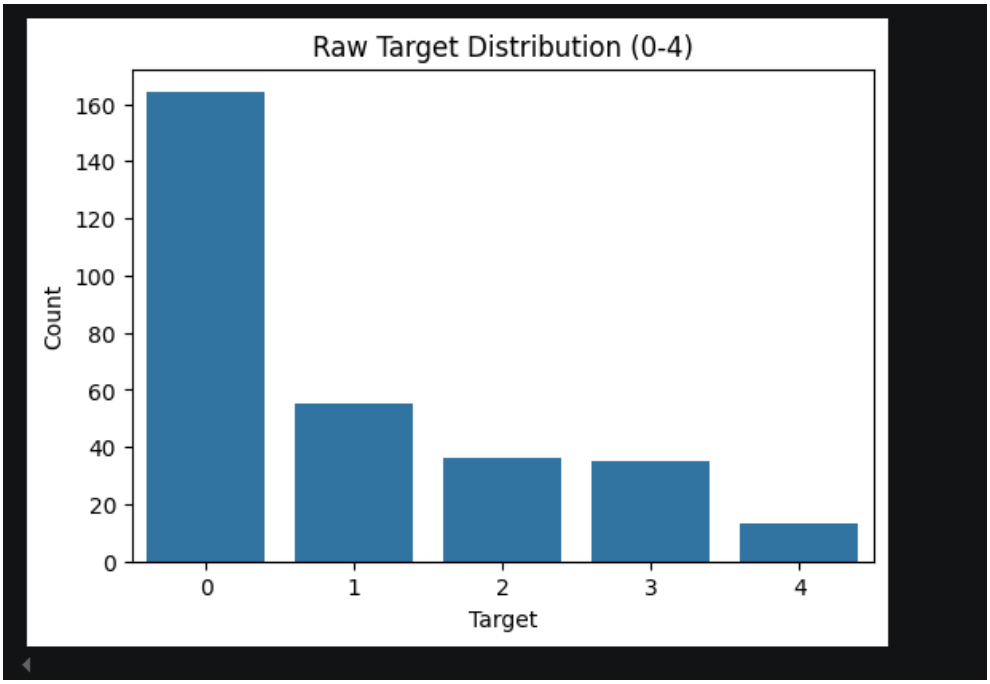
Balanced vs imbalanced data

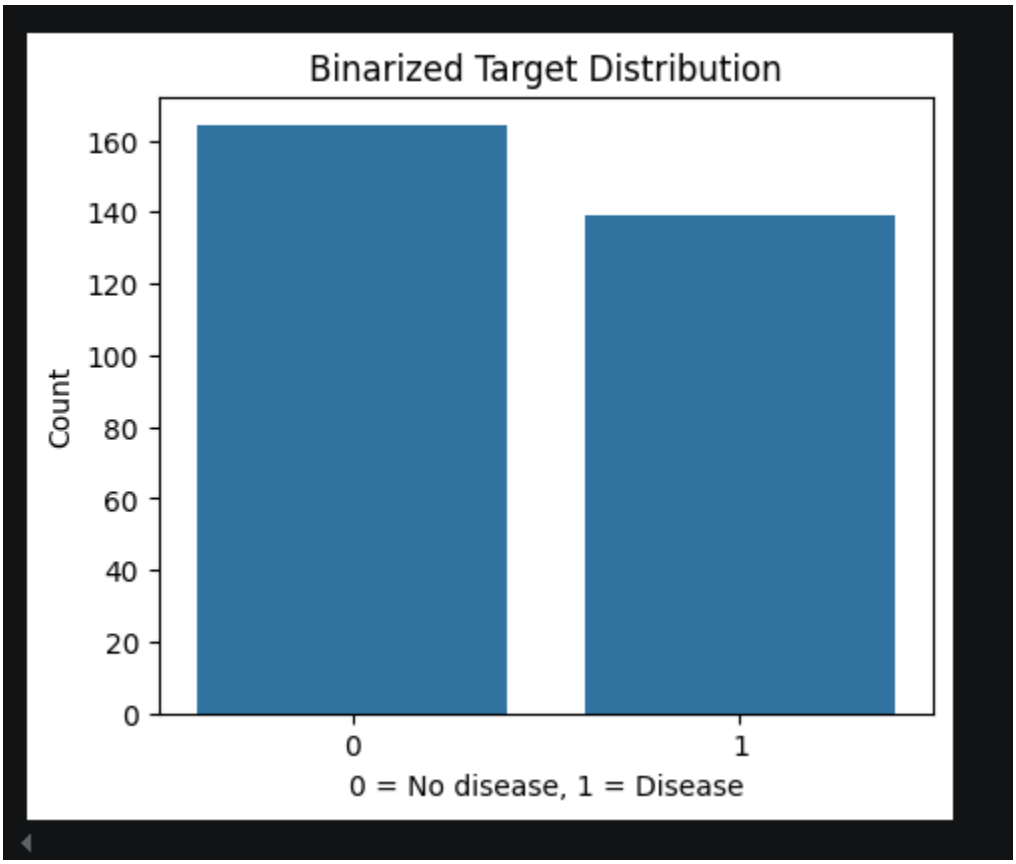
```
Raw target class counts (0 = no disease, 1-4 = increasing severity):
target
0    164
1     55
2     36
3     35
4     13
Name: count, dtype: int64

Raw target class percentages:
target
0    54.13
1    18.15
2    11.88
3    11.55
4     4.29
Name: proportion, dtype: float64

Binarized class counts (0 = no disease, 1 = disease):
target
0    164
1    139
Name: count, dtype: int64

Binarized class percentages:
target
0    54.13
1    45.87
Name: count, dtype: float64
```


Raw Target Distribution (0-4)
Count
160
140
120
100
80
60
40
20
0
0
1
2
3
4
Target


Binarized Target Distribution
Count
160
140
120
100
80
60
40
20
0
0
1
0 = No disease, 1 = Disease

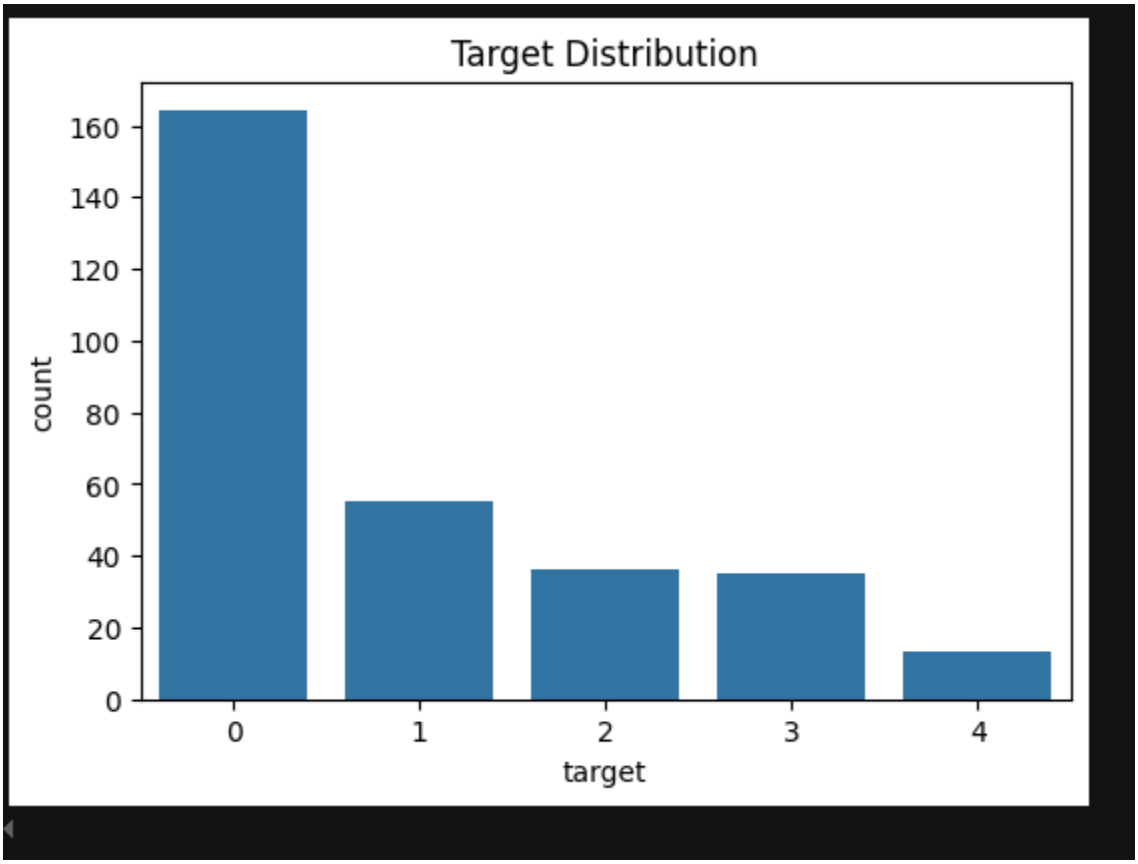
```
Missing values per column:
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          4
thal        2
target      0
dtype: int64

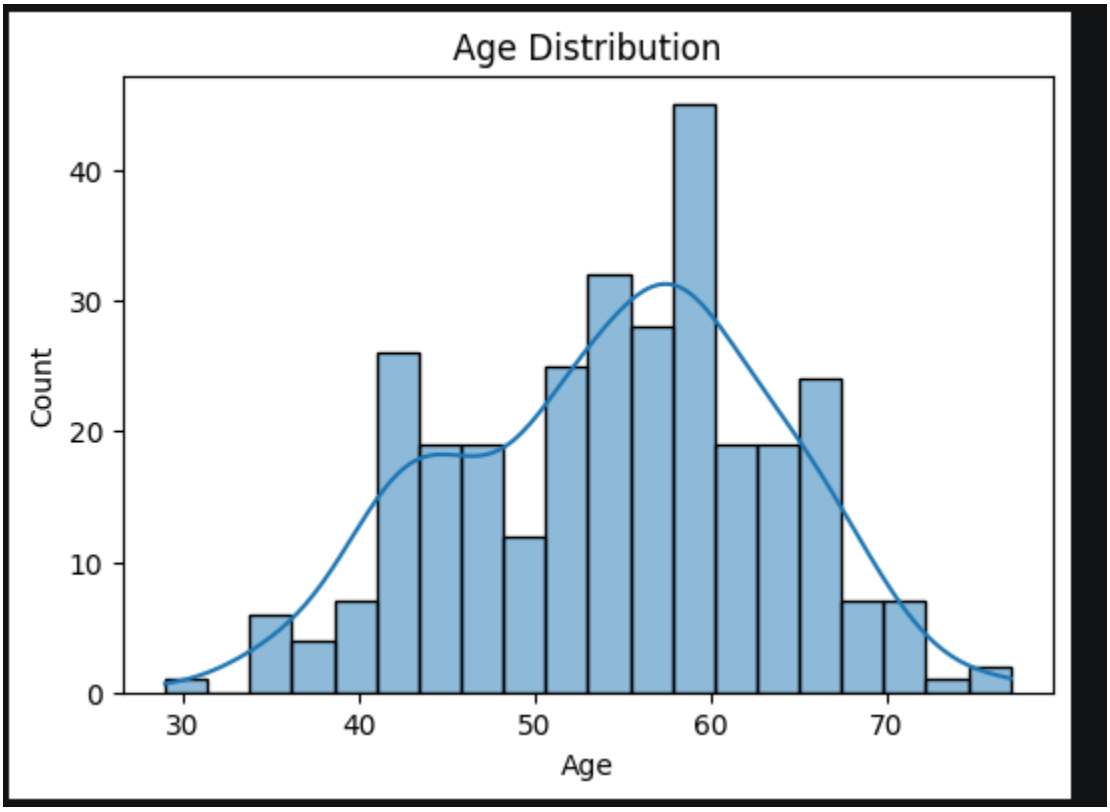
Total missing values in dataset: 6

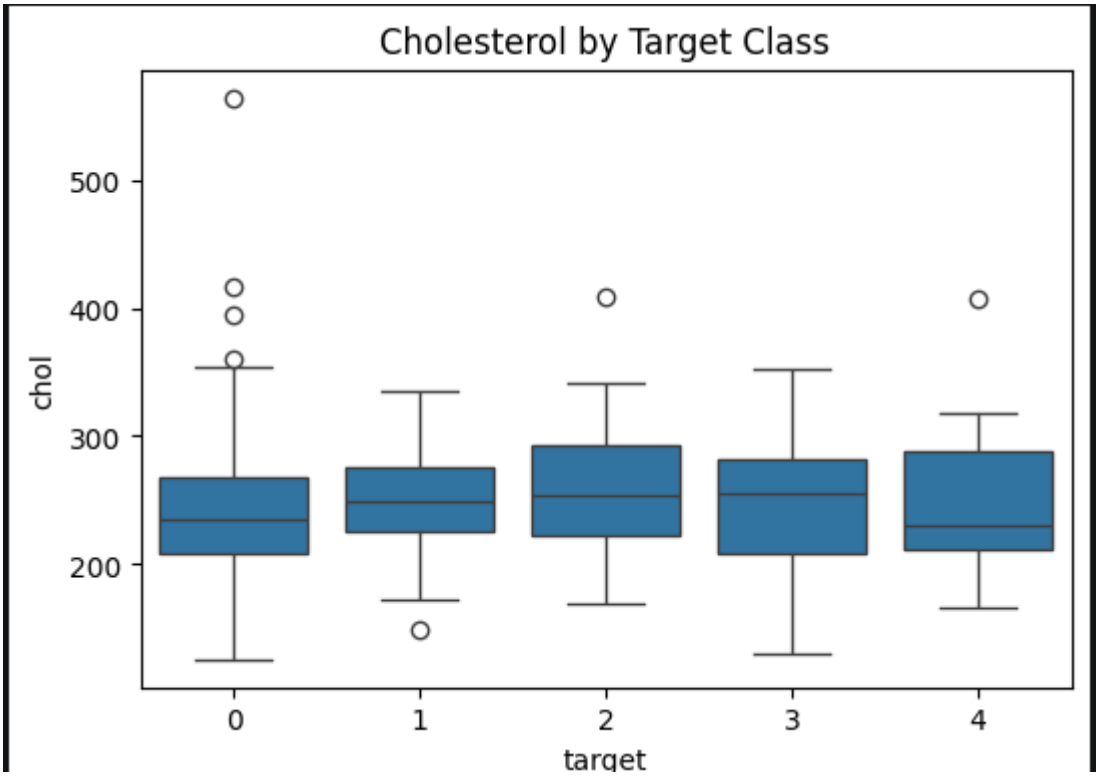
Duplicate rows: 0

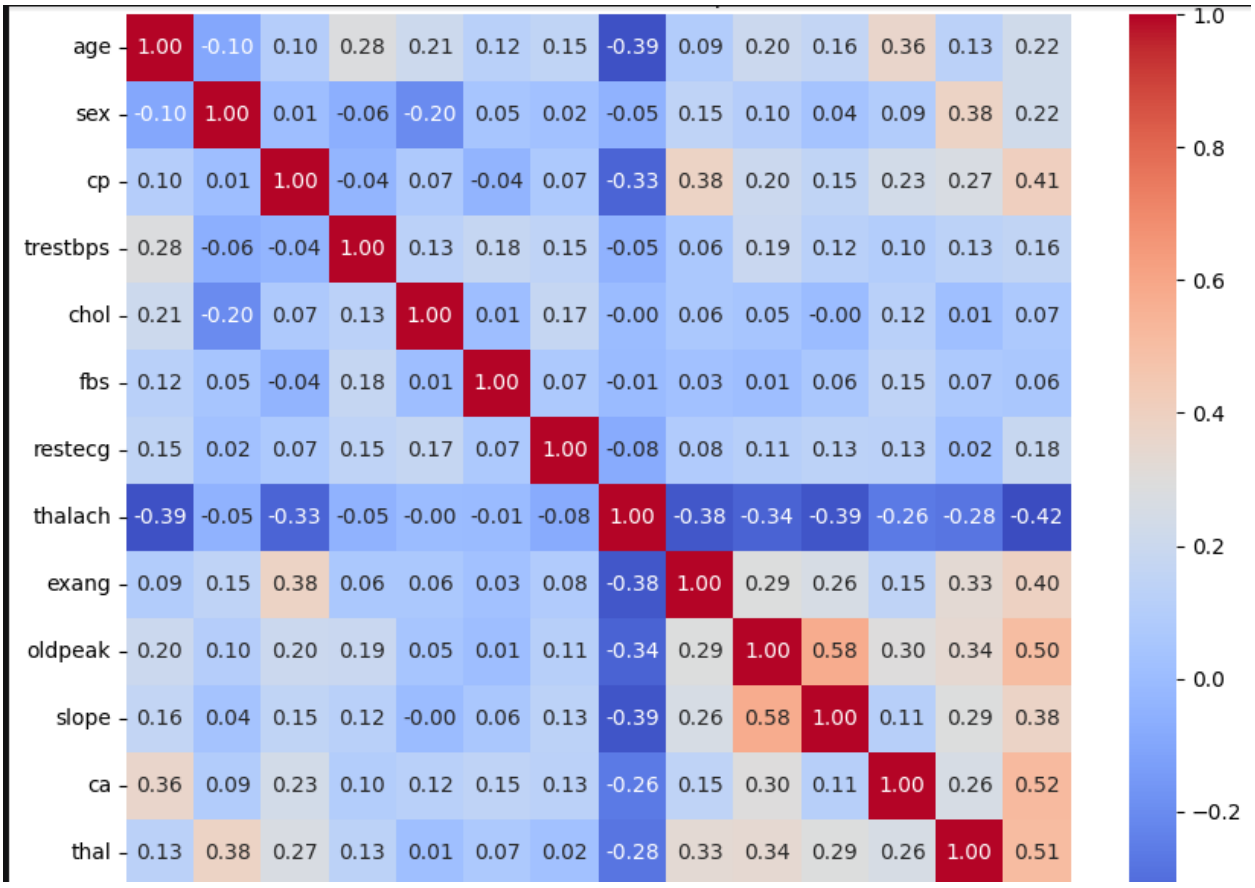
Data types:
age         float64
sex         float64
cp          float64
...
ca            4
thal          3
target        5
dtype: int64
```

Visualize data

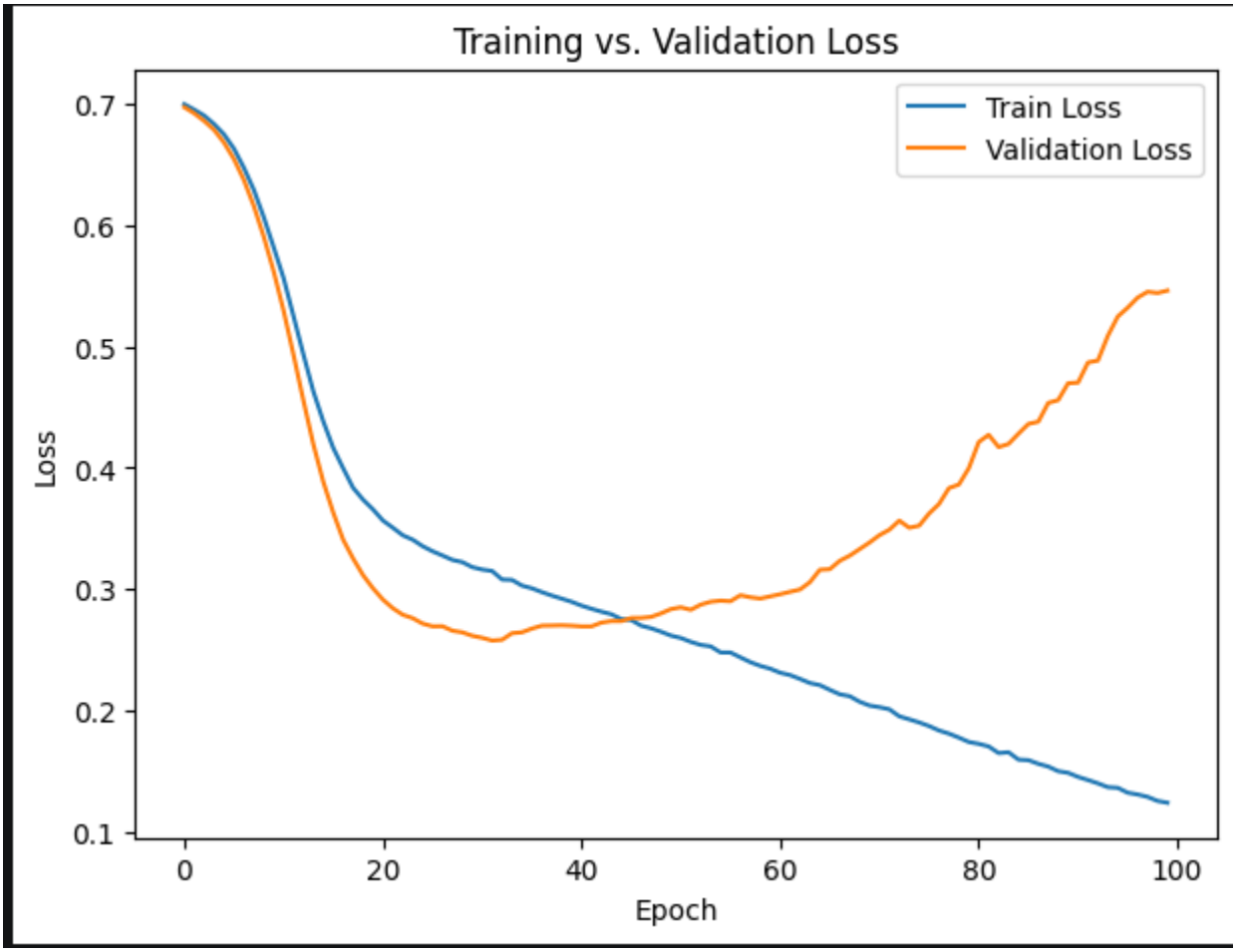

Target Distribution
count
target
0
1
2
3
4
0
20
40
60
80
100
120
140
160


Age Distribution
Count
Age
30
40
50
60
70
0
10
20
30
40


Cholesterol by Target Class
chol
500
400
300
200
0
1
2
3
4
target


age 1.00 -0.10 0.10 0.28 0.21 0.12 0.15 -0.39 0.09 0.20 0.16 0.36 0.13 0.22
sex -0.10 1.00 0.01 -0.06 -0.20 0.05 0.02 -0.05 0.15 0.10 0.04 0.09 0.38 0.22
cp 0.10 0.01 1.00 -0.04 0.07 -0.04 0.07 -0.33 0.38 0.20 0.15 0.23 0.27 0.41
trestbps 0.28 -0.06 -0.04 1.00 0.13 0.18 0.15 -0.05 0.06 0.19 0.12 0.10 0.13 0.16
chol 0.21 -0.20 0.07 0.13 1.00 0.01 0.17 -0.00 0.06 0.05 -0.00 0.12 0.01 0.07
fbs 0.12 0.05 -0.04 0.18 0.01 1.00 0.07 -0.01 0.03 0.01 0.06 0.15 0.07 0.06
restecg 0.15 0.02 0.07 0.15 0.17 0.07 1.00 -0.08 0.08 0.11 0.13 0.13 0.02 0.18
thalach -0.39 -0.05 -0.33 -0.05 -0.00 -0.01 -0.08 1.00 -0.38 -0.34 -0.39 -0.26 -0.28 -0.42
exang 0.09 0.15 0.38 0.06 0.06 0.03 0.08 -0.38 1.00 0.29 0.26 0.15 0.33 0.40
oldpeak 0.20 0.10 0.20 0.19 0.05 0.01 0.11 -0.34 0.29 1.00 0.58 0.30 0.34 0.50
slope 0.16 0.04 0.15 0.12 -0.00 0.06 0.13 -0.39 0.26 0.58 1.00 0.11 0.29 0.38
ca 0.36 0.09 0.23 0.10 0.12 0.15 0.13 -0.26 0.15 0.30 0.11 1.00 0.26 0.52
thal 0.13 0.38 0.27 0.13 0.01 0.07 0.02 -0.28 0.33 0.34 0.29 0.26 1.00 0.51
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

```
Epoch 10/100 | Train Loss: 0.5820 | Val Loss: 0.5621
Epoch 20/100 | Train Loss: 0.3658 | Val Loss: 0.3007
Epoch 30/100 | Train Loss: 0.3181 | Val Loss: 0.2615
Epoch 40/100 | Train Loss: 0.2896 | Val Loss: 0.2698
Epoch 50/100 | Train Loss: 0.2617 | Val Loss: 0.2835
Epoch 60/100 | Train Loss: 0.2344 | Val Loss: 0.2940
Epoch 70/100 | Train Loss: 0.2040 | Val Loss: 0.3384
Epoch 80/100 | Train Loss: 0.1740 | Val Loss: 0.3997
Epoch 90/100 | Train Loss: 0.1485 | Val Loss: 0.4698
Epoch 100/100 | Train Loss: 0.1239 | Val Loss: 0.5462
```



```
Accuracy : 0.8689
Precision: 0.8333
Recall   : 0.8929
F1-score : 0.8621
```


```
Accuracy : 0.8689
Precision: 0.8333
Recall   : 0.8929
F1-score : 0.8621
```


	Model	Accuracy	Precision	Recall	F1-Score
0	Neural Network	0.868852	0.833333	0.892857	0.862069
1	AdaBoost	0.868852	0.833333	0.892857	0.862069

Cell 20 — Final Interpretation

1. **Balance:** The binarized target (disease vs. no disease) splits 164 vs. 139 patients (54.13% vs. 45.87%), so the dataset is only mildly imbalanced — no resampling was needed, but recall was still worth checking alongside accuracy.
2. **Preprocessing:** Missing values appeared only in `ca` (4) and `thal` (2) — 6 total, out of 303 rows — and were imputed with the most frequent value since both are categorical-coded columns. Categorical features were one-hot encoded, numerical features were standardized, and the preprocessor was fit only on the training split to avoid data leakage. This expanded the feature set from 13 raw columns to 25 processed input features.
3. **Architecture:** A 25 → 32 → 16 → 8 → 1 fully-connected network with ReLU activations on the hidden layers and a raw logit output, trained with `BCEWithLogitsLoss` and the Adam optimizer.
4. **Neural network results:** Accuracy = 0.8689, Precision = 0.8333, Recall = 0.8929, F1 = 0.8621. The loss curves show clear overfitting after around epoch 30–40: training loss kept dropping steadily to 0.1239 by epoch 100, while validation loss bottomed out near 0.26 around epoch 30 and then climbed back up to 0.5462 — a widening gap that's the textbook overfitting signature.
5. **AdaBoost results:** Accuracy = 0.8689, Precision = 0.8333, Recall = 0.8929, F1 = 0.8621.
6. **Comparison:** Both models landed on identical scores across all four metrics on this run. That's plausible given the very small test set (~61 patients), where a handful of prediction flips can easily produce matching fractions between two otherwise different models. It also suggests AdaBoost — a much simpler, lower-variance ensemble — matched a from-scratch neural network's performance without needing early stopping or regularization, which lines up with the overfitting seen in the NN's validation loss.
7. **Possible improvements:** early stopping or dropout to fix the neural network's overfitting past ~epoch 30, k-fold cross-validation instead of a single train/test split (especially given how small the test set is), hyperparameter tuning (hidden layer sizes, learning rate, `n_estimators`), and trying a stronger gradient boosting baseline like XGBoost or LightGBM for a third comparison point.